Network-Flow Games: Hardness of Nucleolus Computation

Felix Zhou

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Cooperative Game Theory

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- Focus: Which coalitions will form and the resulting collective payoffs.
- High-level approach describing structure, strategies, and payoffs. Ignore division of payoffs between coalitions.

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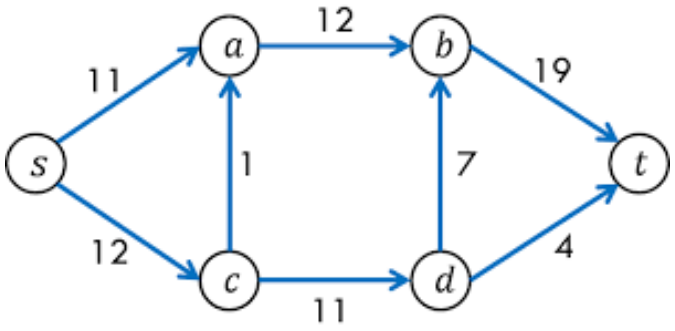
- *Allocation:* $x \in \mathbb{R}^n : x(N) = \nu(N)$.
- *Imputation:*
  $X(\Gamma) := \{x \in \mathbb{R}^n : x(N) = v(N) \wedge \forall i \in N, x(i) \geq \nu(\{i\})\}$.

Network Flow Games

- Instance of maximum flow: Directed graph $G = (V, E)$, source/sink s, t, and capacities $c : E \to \mathbb{R}_{\geq 0}$.



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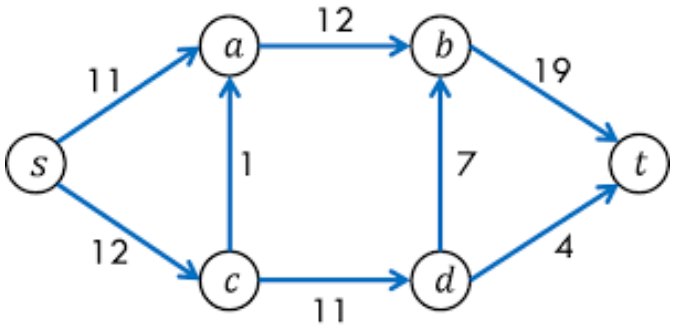

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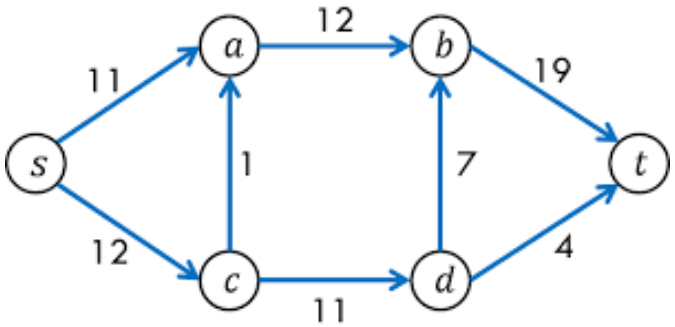
- Characteristic Function: For $S \subseteq E$, $\nu(S)$ denotes maximum flow in $G[S]$.



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- $\Theta(x) \in \mathbb{R}^{2n-2}$: entries are $e(S, x), \varnothing \neq S \subset N$, sorted in non-decreasing order.
- *Nucleolus:* (unique) imputation maximizing $\Theta(x)$ lexicographically.

Maschler's Scheme

- Idea: Compute the nucleolus by solving a sequence of linear programs $LP_k, k \geq 1$.

$$
\begin{aligned}
LP_k : \max \epsilon & \\
x(S) = \nu(S) + \epsilon_r & \qquad \forall S \in \mathcal{J}_r, 0 \leq r \leq k-1 \\
x(S) \geq \nu(S) + \epsilon & \qquad \forall S \in \mathcal{J} \setminus \bigcup_{r=0}^{k-1} \mathcal{J}_r
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- $\mathcal{J}_r := \{S \subseteq N : x(S) = \nu(S) + \epsilon_r, (x, \epsilon_r) \in X_r\}, r \geq 1$ (tight coalitions).

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- Take
  $\mathcal{J} := \{\{e\} : e \in E\} \cup \{F \subseteq E : \text{union of } st\text{-dipaths in } G\}$.

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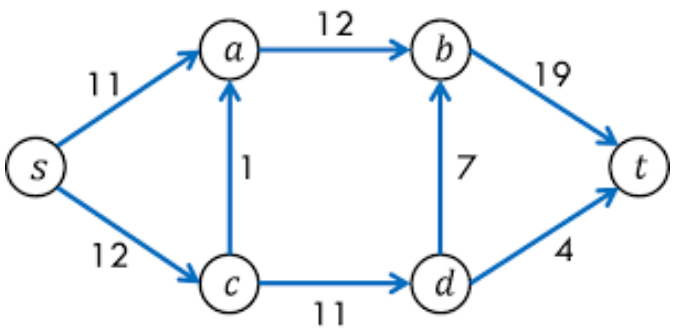
Hardness of Nucleolus Testing

Theorem [Deng, Fang, et al. '09]

Computing the nucleolus of a flow game Γ with integral arc capacities is $\mathcal{NP}$-hard.



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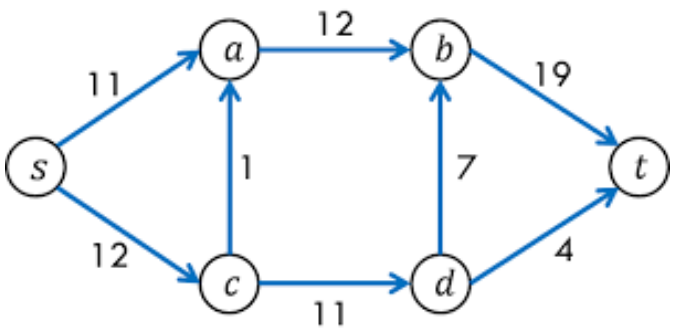
Checking whether an allocation $y \in C(\Gamma)$ is the nucleolus of a flow game Γ with integral arc capacities is $\mathcal{NP}$-hard.



Other Results

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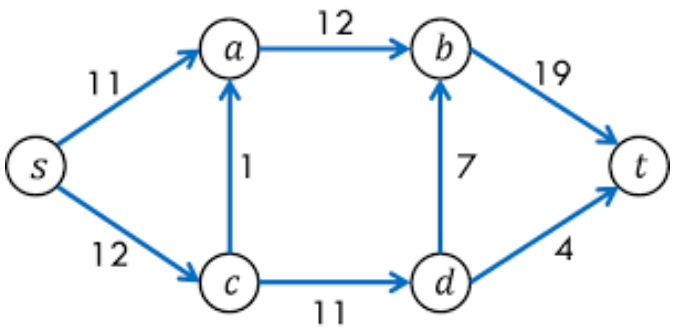
The nucleolus of a *simple* flow game Γ is polynomial-time computable.



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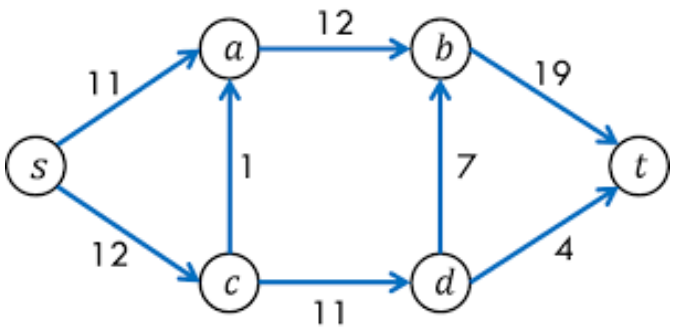
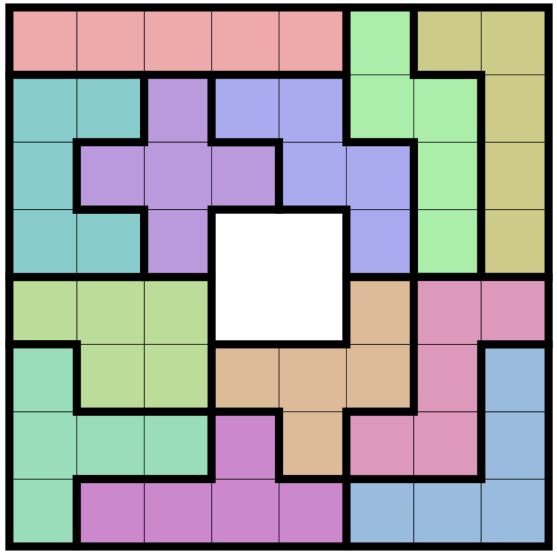
Theorem [Potters, Reijinierse, et al. '06]

There is a polynomial-time combinatorial algorithm which computes the nucleolus of a simple flow game Γ, assuming $C(\Gamma) \neq \varnothing$.



Exact Cover

- Finite set
  $U = \{u_1, u_2, \ldots, u_{3q}\}, q \geq 1.$



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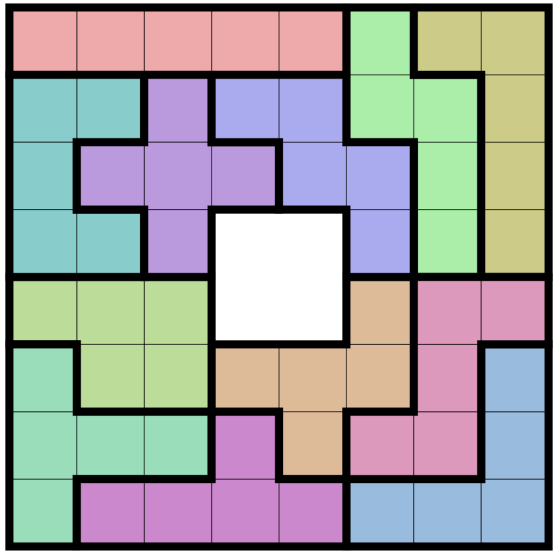

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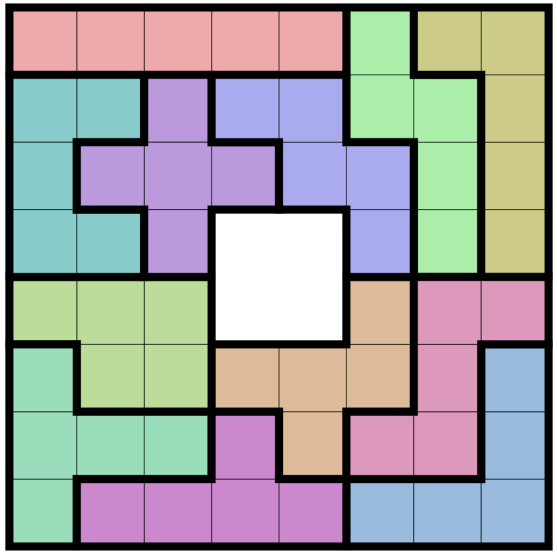

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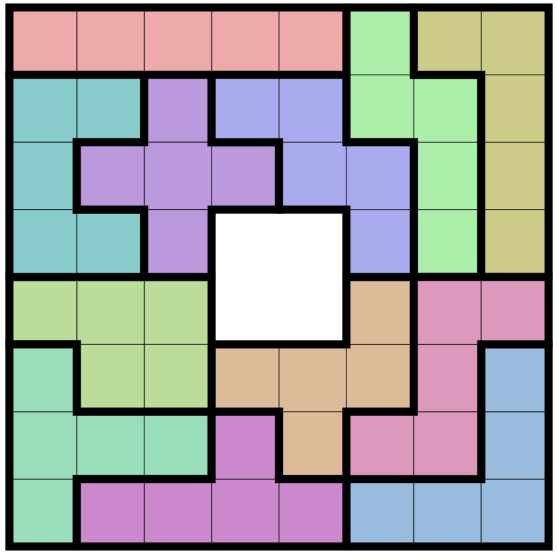
- One of Karps's 21 $\mathcal{NP}$-complete problems.



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- $\mathcal{NP}$-complete [Garey & Johnson, '79].

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u'_1 u'_2 u'_3

u'_6 u'_4

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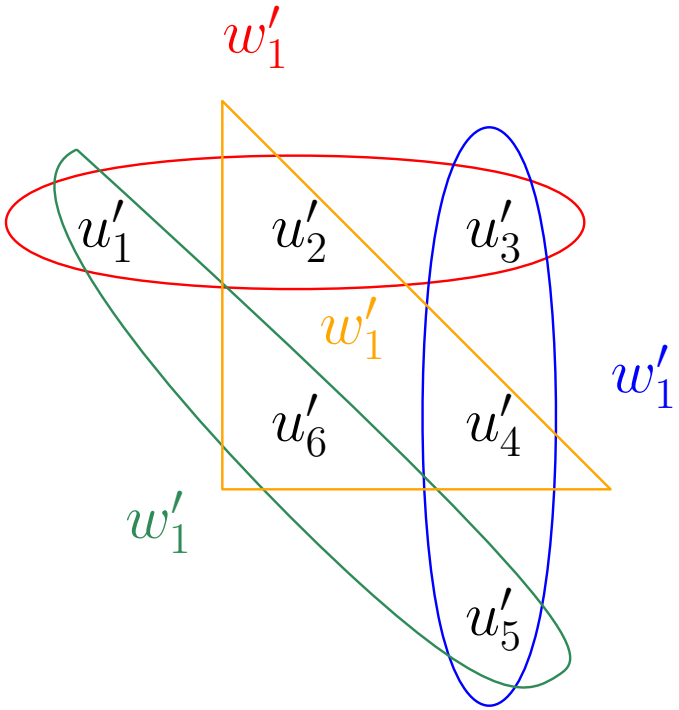

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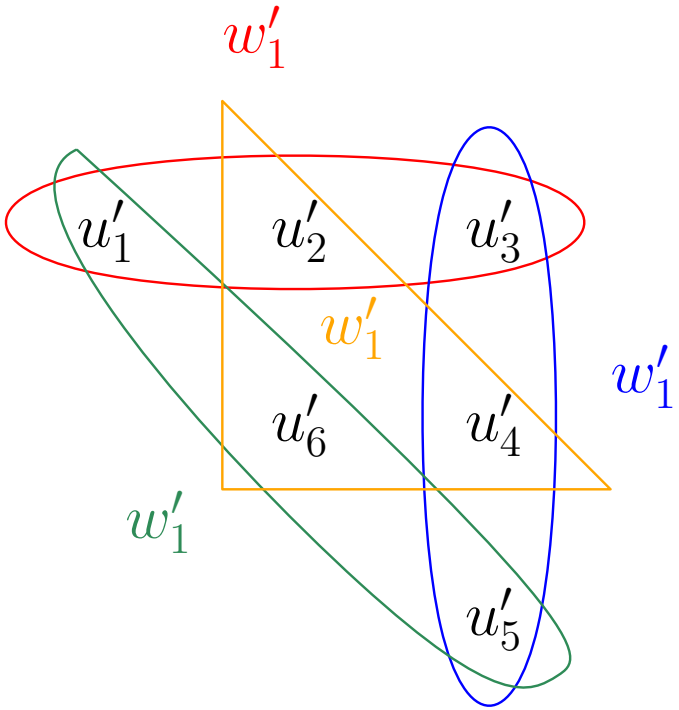

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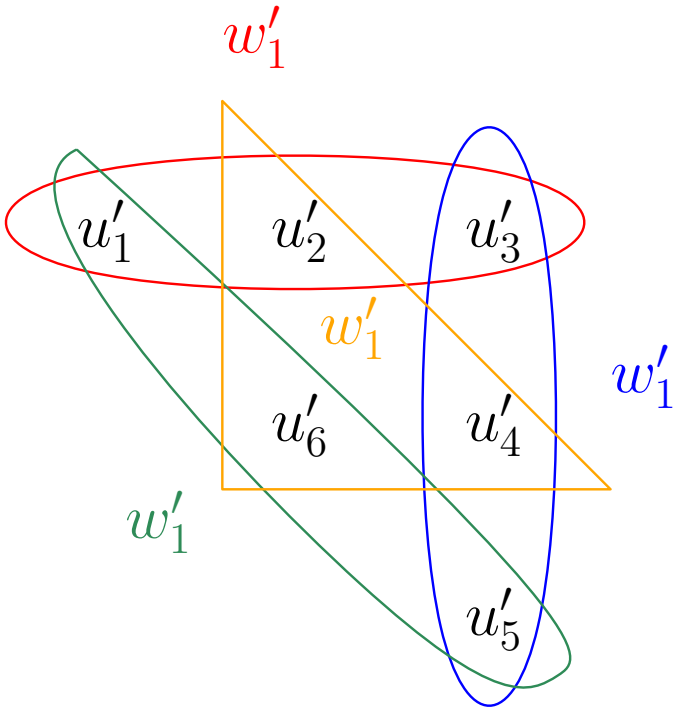

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- Construct specific allocation which is the nucleolus if and only if network comes from "YES" instance of X3C.

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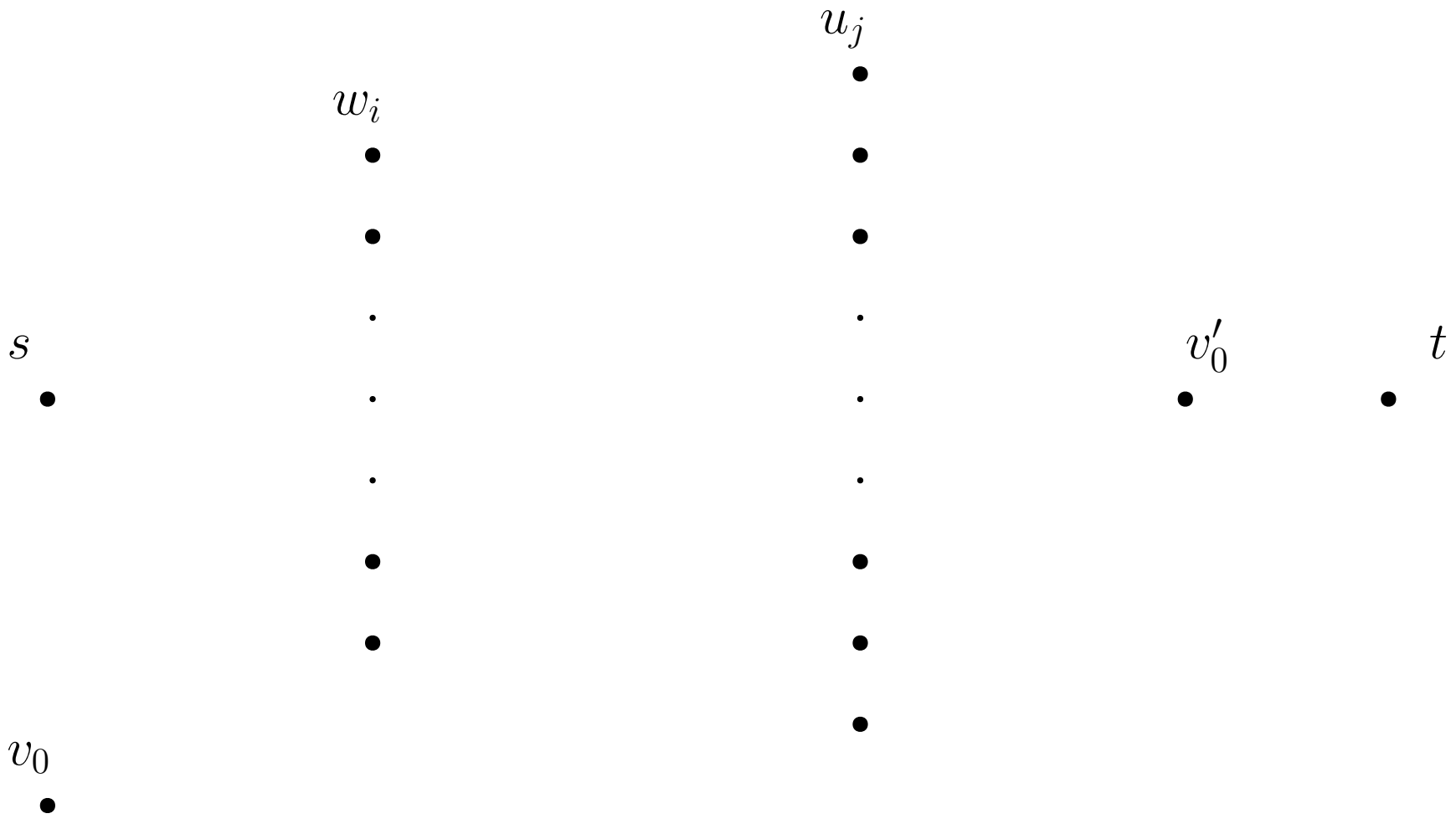
- $|\bar{U}| = 3\bar{q} = 3(q + 6), |\bar{W}| = \bar{f} = f + 12$
- Minimum cover K has size $|K| = \bar{q} + 3$ if and only if original instance has exact cover.

Constructed Network



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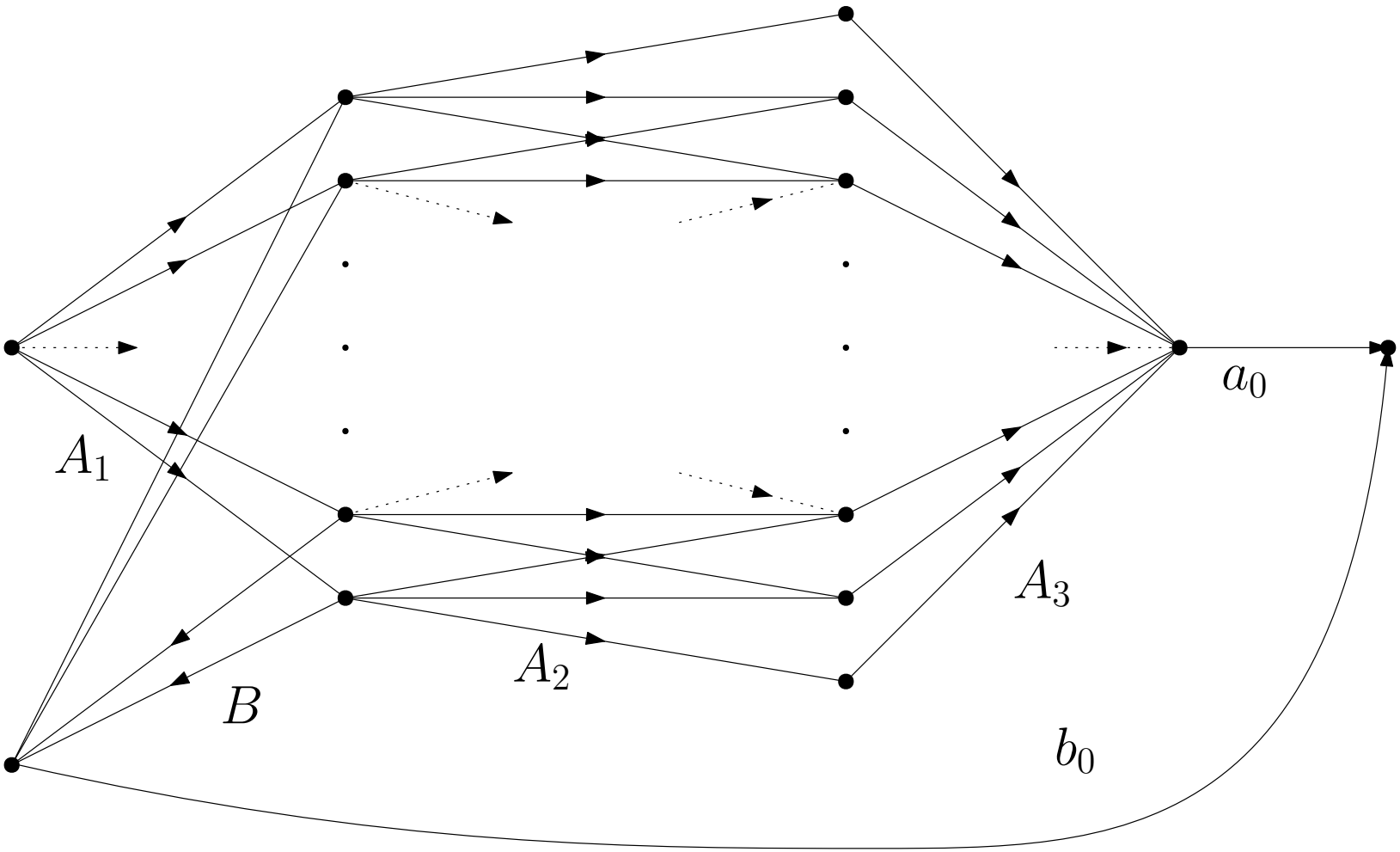

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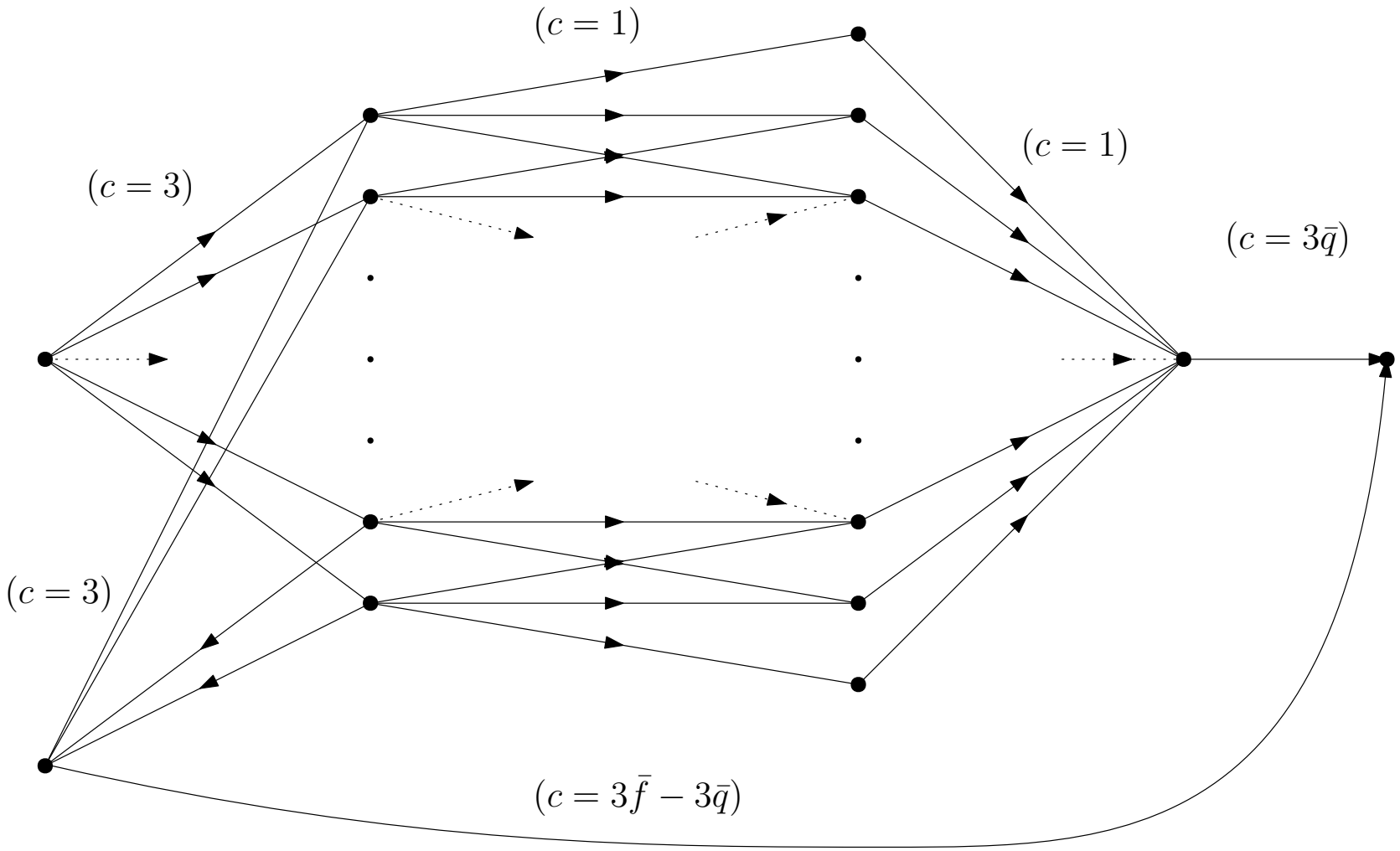

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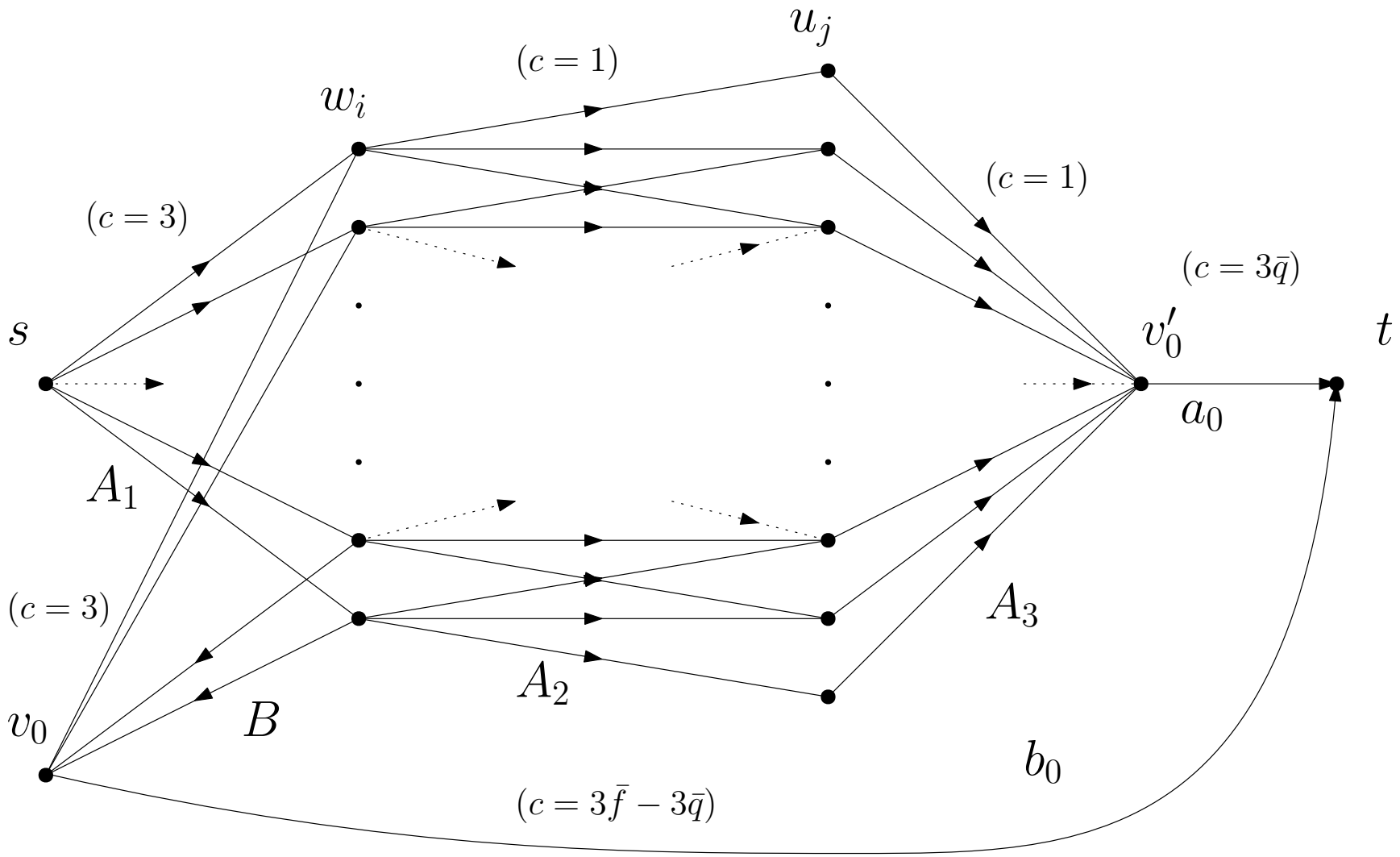
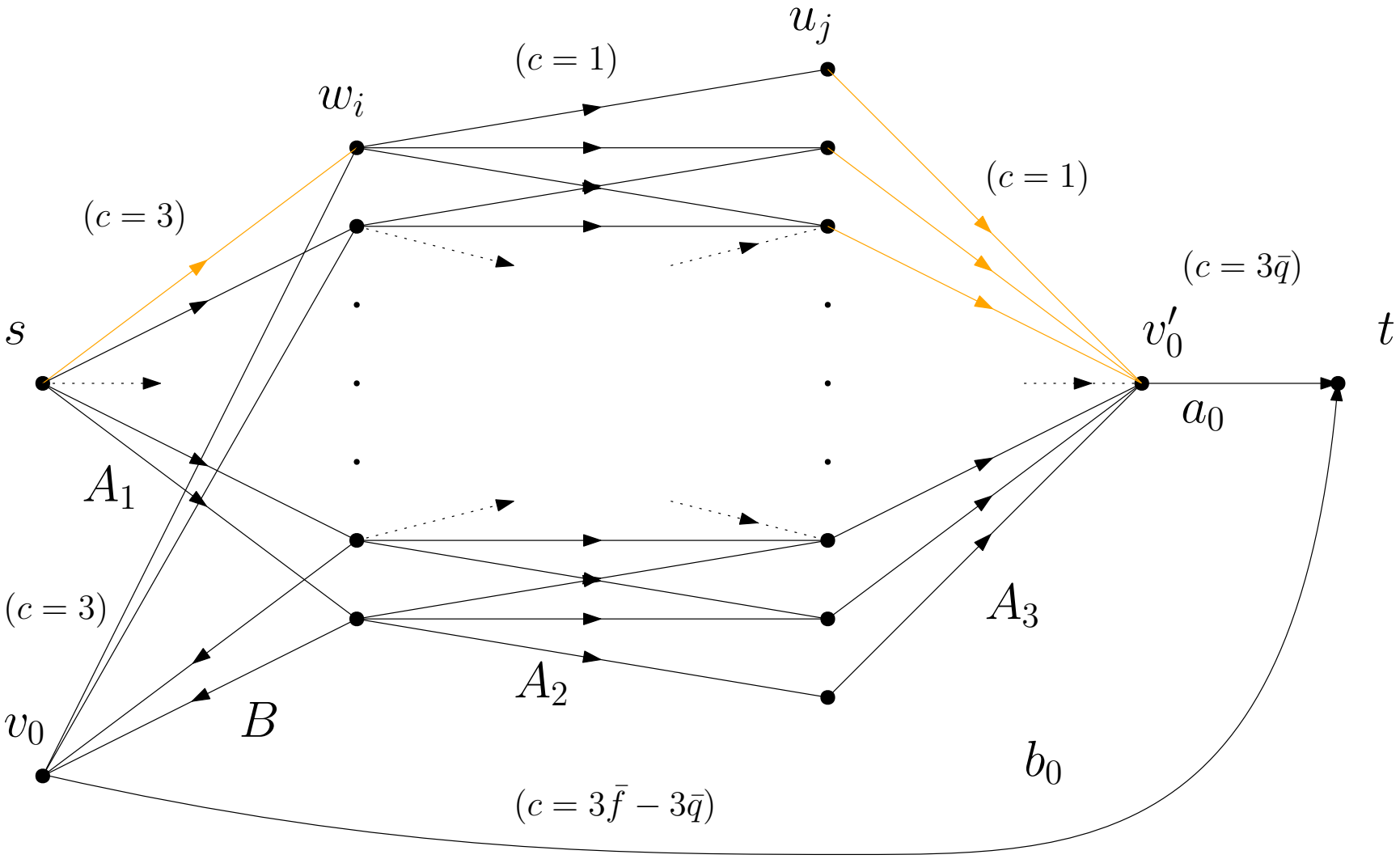


Constructed Network

$T(e)$ for $e \in A_1$.



Some Lower Bounds

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Cases	$L(3\bar{q}) = \bar{q} + 3$	$L(3\bar{q}) \geq \bar{q} + 4$
$k = 3\bar{q} - 1$	$\bar{q} + 2$	$\bar{q} + 3$
$k = 3\bar{q} - 2$	$\bar{q} + 1$	$\bar{q} + 2$
$k = 3\bar{q} - 3, 3\bar{q} - 4$	$\bar{q}$	$\bar{q} + 1$
$k = 3\bar{q} - 5, 3\bar{q} - 6$	$\bar{q} - 1$	$\bar{q}$
$k \leq 3\bar{q} - 7$	$\left\lceil \frac{k}{3} \right\rceil$	$\left\lceil \frac{k}{3} \right\rceil$

Table: Lower bounds for $L(k)$

A Specific Allocation

$$\eta_\delta(e) := \begin{cases} \frac{3}{4}, & e \in A_1 \\ 0, & e \in A_2 \cup B \\ \frac{1}{2} & e \in A_3 \\ \frac{3}{4}\bar{q} - \frac{3}{4} - \delta, & e = a_0 \\ \frac{9}{4}(\bar{f} - \bar{q}) + \frac{3}{4} + \delta, & e = b_0 \end{cases}$$

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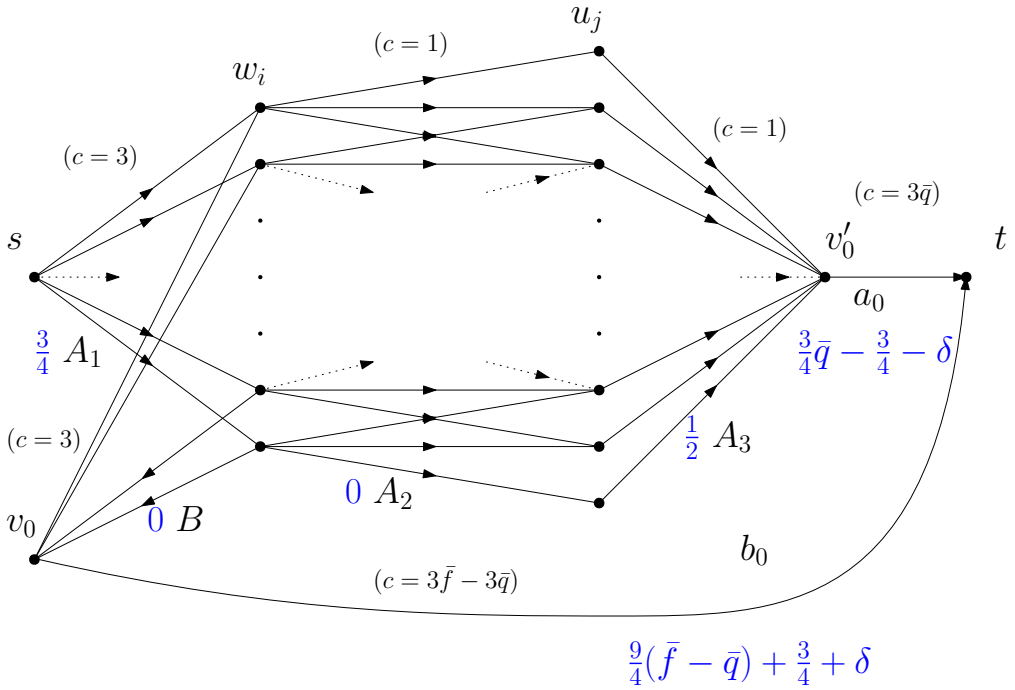


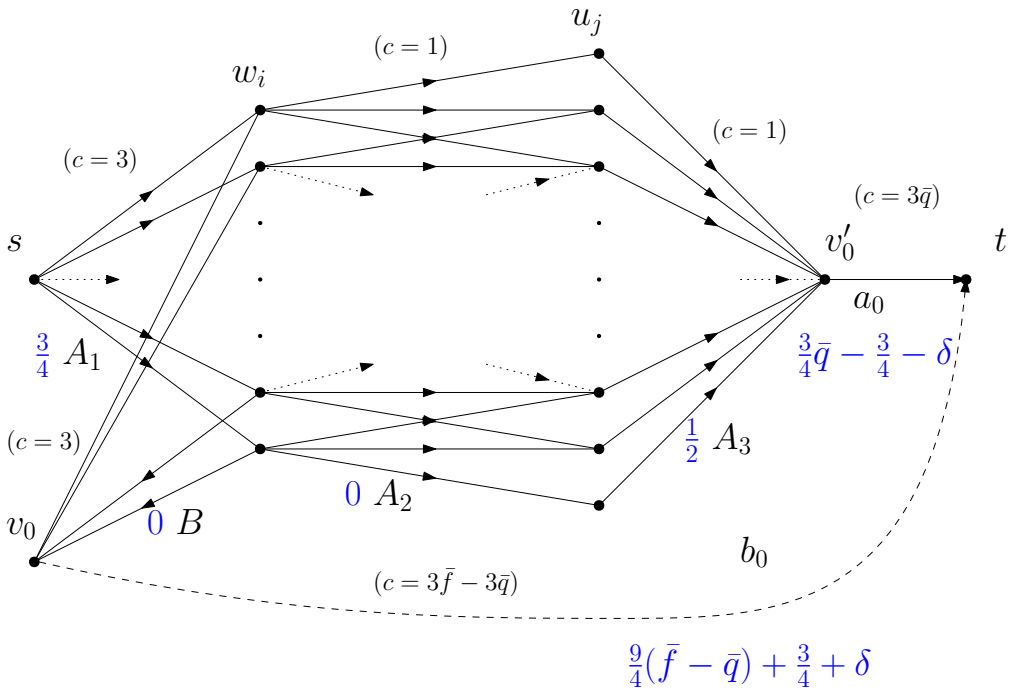
Excess Lemma

Lemma (Excess)

Let $S \in \mathcal{J} \setminus \mathcal{J}_0$ with $\nu(S) > 0$.

1. If $a_0 \in S, b_0 \notin S$

   Case $L(3\bar{q}) = \bar{q} + 3$: $\eta_\delta(S) - \nu(S) \geq \frac{3}{4} - \delta$.



Excess Lemma

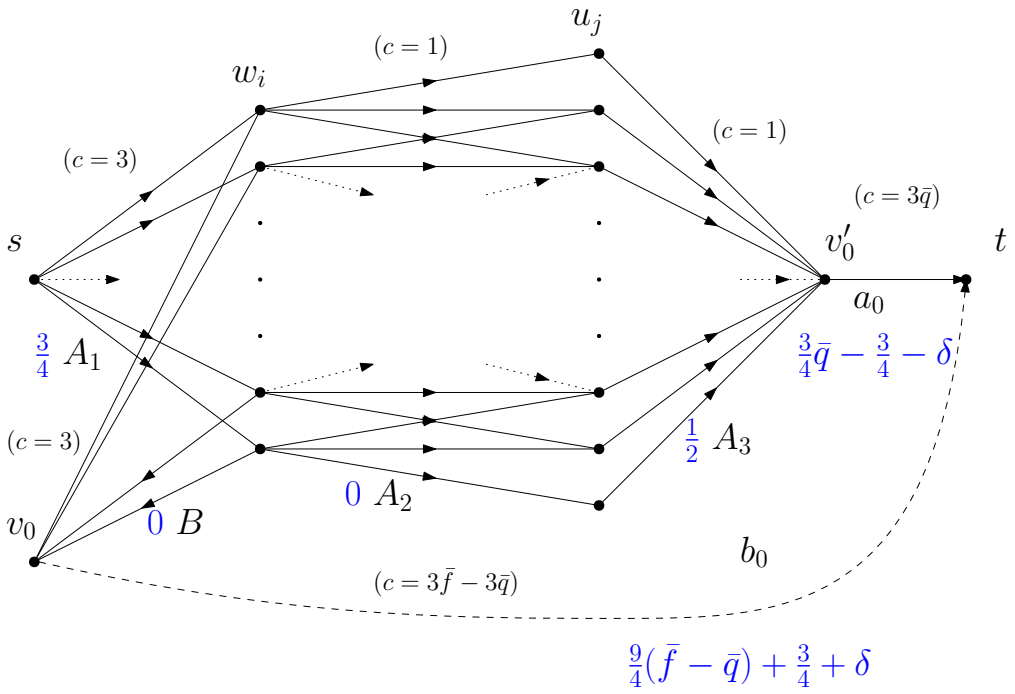
Lemma (Excess)

Let $S \in \mathcal{J} \setminus \mathcal{J}_0$ with $\nu(S) > 0$.

1. If $a_0 \in S, b_0 \notin S$

   Case $L(3\bar{q}) = \bar{q} + 3$: $\eta_\delta(S) - \nu(S) \geq \frac{3}{4} - \delta$.

   Case $L(3\bar{q}) \geq \bar{q} + 4$: $\eta_\delta(S) - \nu(S) \geq 1 - \delta$.



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Proof: If S has $\nu(S) = k$ arcs of A_3,
it must have at least $L(k)$ arcs of A_1.

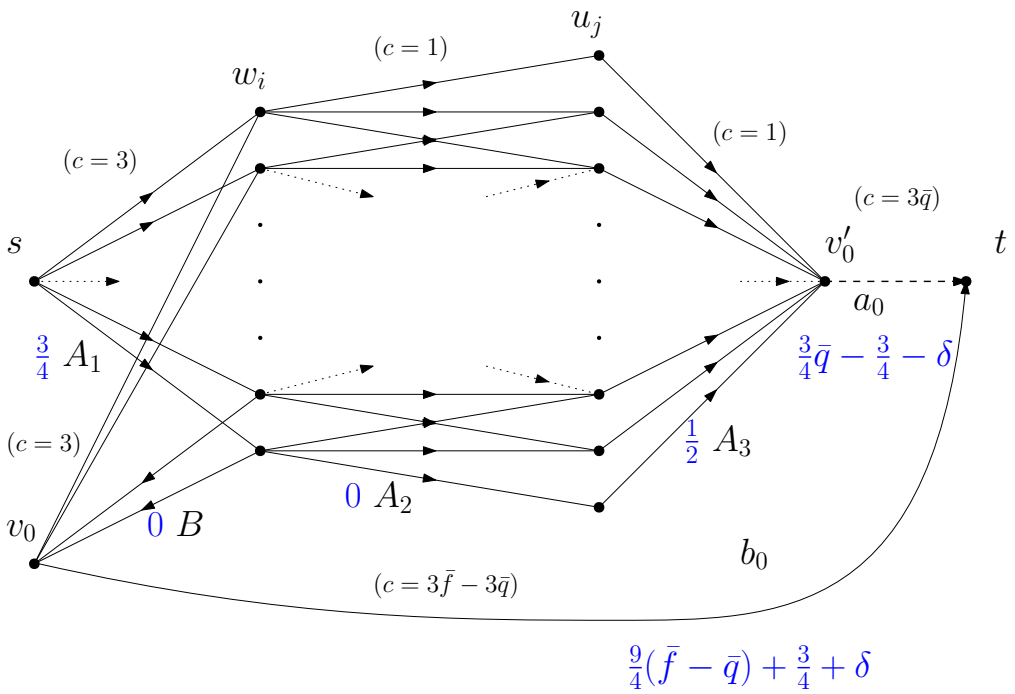
$$\eta_\delta(S) - \nu(S) \geq \left[\left(\frac{3}{4}\bar{q} - \frac{3}{4} - \delta\right) + \frac{1}{2}k + \frac{3}{4}L(k)\right] - k$$

Excess Lemma

Lemma (Excess)

Let $S \in \mathcal{J} \setminus \mathcal{J}_0$ with $\nu(S) > 0$.

2. If $a_0 \notin S, b_0 \in S$, $\eta_\delta(S) - \nu(S) \geq \frac{3}{4} + \delta$.



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2. If $a_0 \notin S, b_0 \in S$, $\eta_\delta(S) - \nu(S) \geq \frac{3}{4} + \delta$.

Proof: The subset minimizing $\eta_\delta(S) - v(S)$ contains $\bar{f} - \bar{q}$ arcs of A_1 and $\nu(S) = 3\bar{f} - 3\bar{q}$.

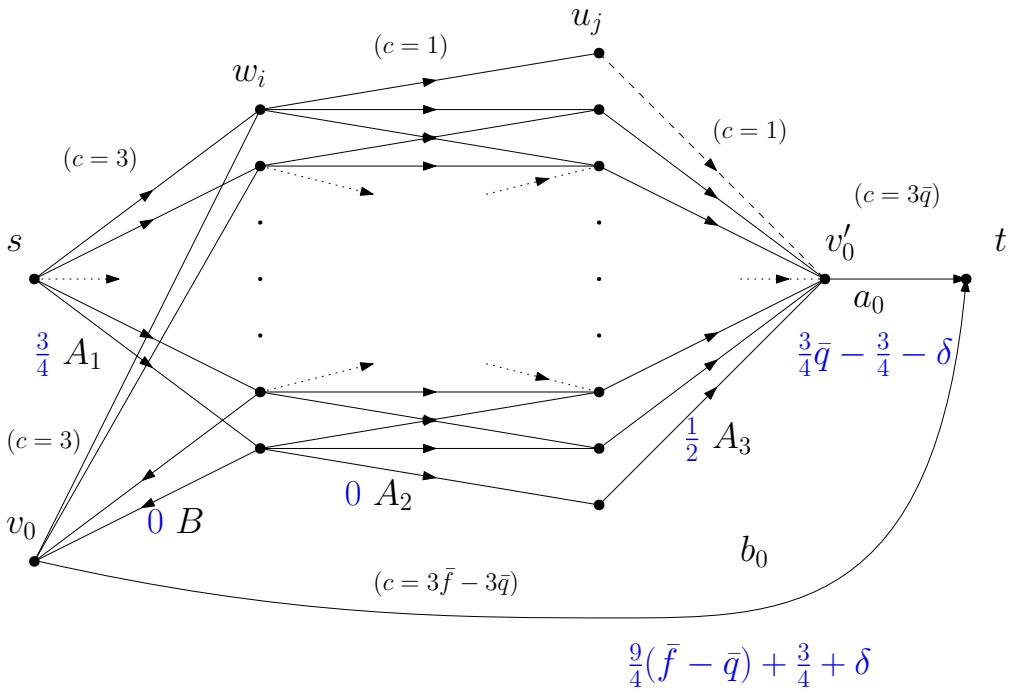
$$\eta_\delta(S) - \nu(S) = \eta_\delta(b_0) + \frac{3}{4}(\bar{f} - \bar{q}) - (3\bar{f} - 3\bar{q}) = \frac{3}{4} + \delta.$$

Excess Lemma

Lemma (Excess)

Let $S \in \mathcal{J} \setminus \mathcal{J}_0$ with $\nu(S) > 0$.

3. Suppose $a_0, b_0 \in S$. Let $\mathcal{S}_1 := \{E \setminus \{e\} : e \in A_3\}$ and $\mathcal{S}_2 := \{E \setminus T(e) : e \in A_1\}$.
   Case $S \in \mathcal{S}_1$: $\eta_\delta(S) - \nu(S) = \frac{1}{2}$



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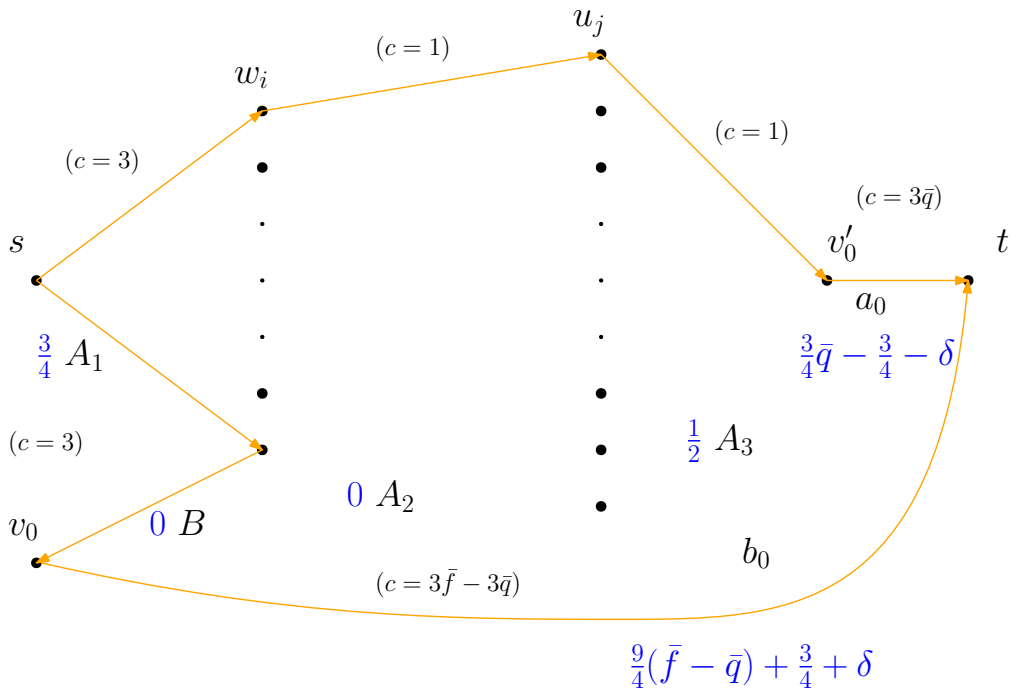
   Case $S \in \mathcal{S}_2$: $\eta_\delta(S) - \nu(S) = \frac{3}{4}$.



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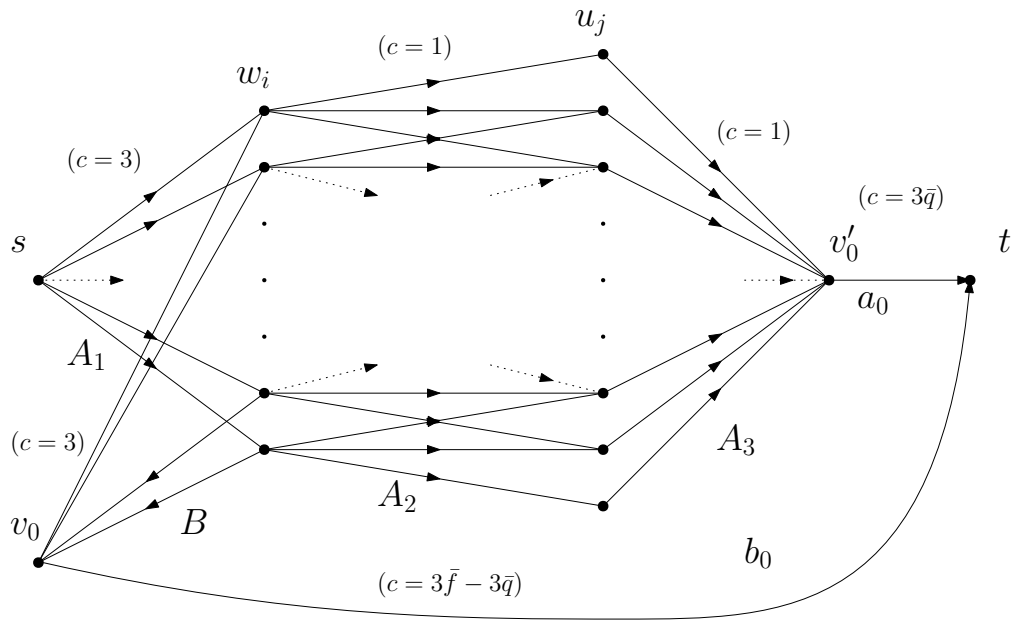
   Case $S \notin \mathcal{S}_1 \cup \mathcal{S}_2$: $\eta_\delta(S) - \nu(S) > \frac{3}{4}$.



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Proof: By calculation,

$$\eta_\delta(S) - \nu(S) \begin{cases} = \frac{1}{2}, & S \in \mathcal{S}_1 \\ = \frac{3}{4}, & S \in \mathcal{S}_2 \\ > \frac{3}{4}, & S \notin \mathcal{S}_1 \cup \mathcal{S}_2 \end{cases}$$

Excess Lemma

Lemma (Excess)

Let $S \in \mathcal{J} \setminus \mathcal{J}_0$ with $\nu(S) > 0$.

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   Case $S \in \mathcal{S}_1$: $\eta_\delta(S) - \nu(S) = \frac{1}{2}$

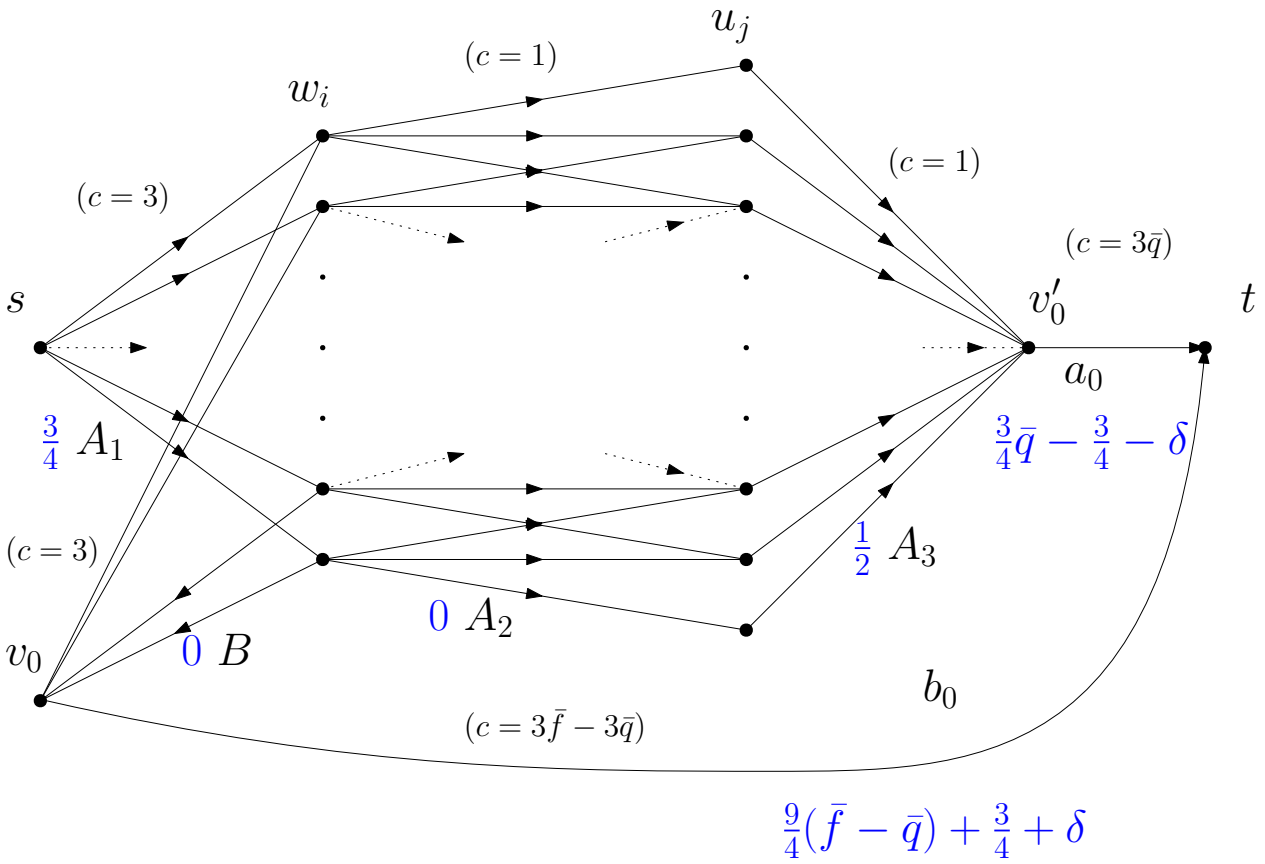
   Case $S \in \mathcal{S}_2$: $\eta_\delta(S) - \nu(S) = \frac{3}{4}$.

   Case $S \notin \mathcal{S}_1 \cup \mathcal{S}_2$: $\eta_\delta(S) - \nu(S) > \frac{3}{4}$.

Maschler's Scheme: First Iteration

Lemma (LP_1)

1. $\eta_\delta \in C(\Gamma)$.



Maschler's Scheme: First Iteration

Lemma (LP_1)

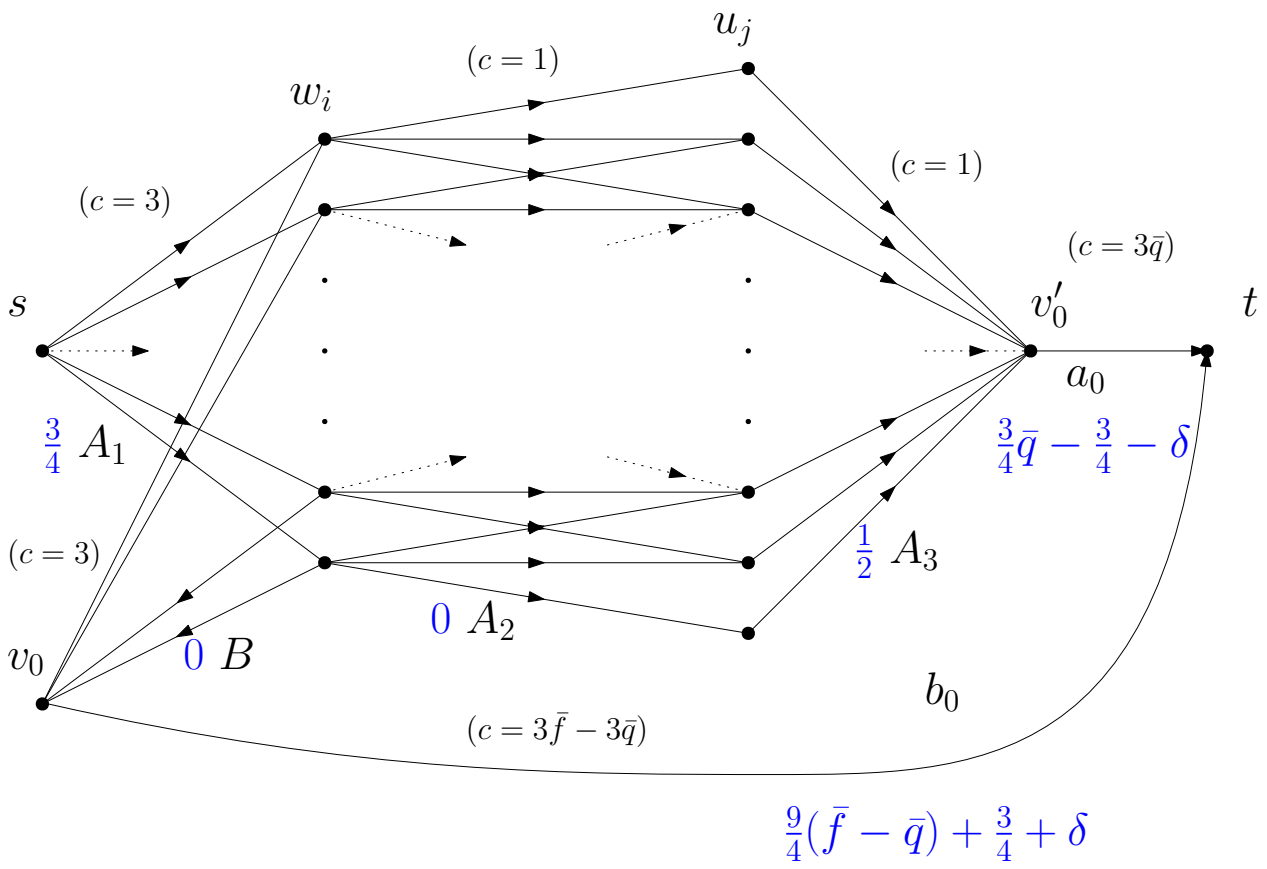
1. $\eta_\delta \in C(\Gamma)$.

Proof: Excess Lemma

Maschler's Scheme: First Iteration

Lemma (LP_1)

(2) $\forall x \in C(\Gamma), x(e) = 0$ for all $e \in A_2 \cup B$.



Maschler's Scheme: First Iteration

Lemma (LP_1)

(2) $\forall x \in C(\Gamma), x(e) = 0$ for all $e \in A_2 \cup B$.

Proof: Any $u \in \bar{U}$ is connected to at least 2 vertices in $\bar{W}$. Thus $\nu(E) = \nu(E \setminus \{e\})$ for all $e \in A_2$ and

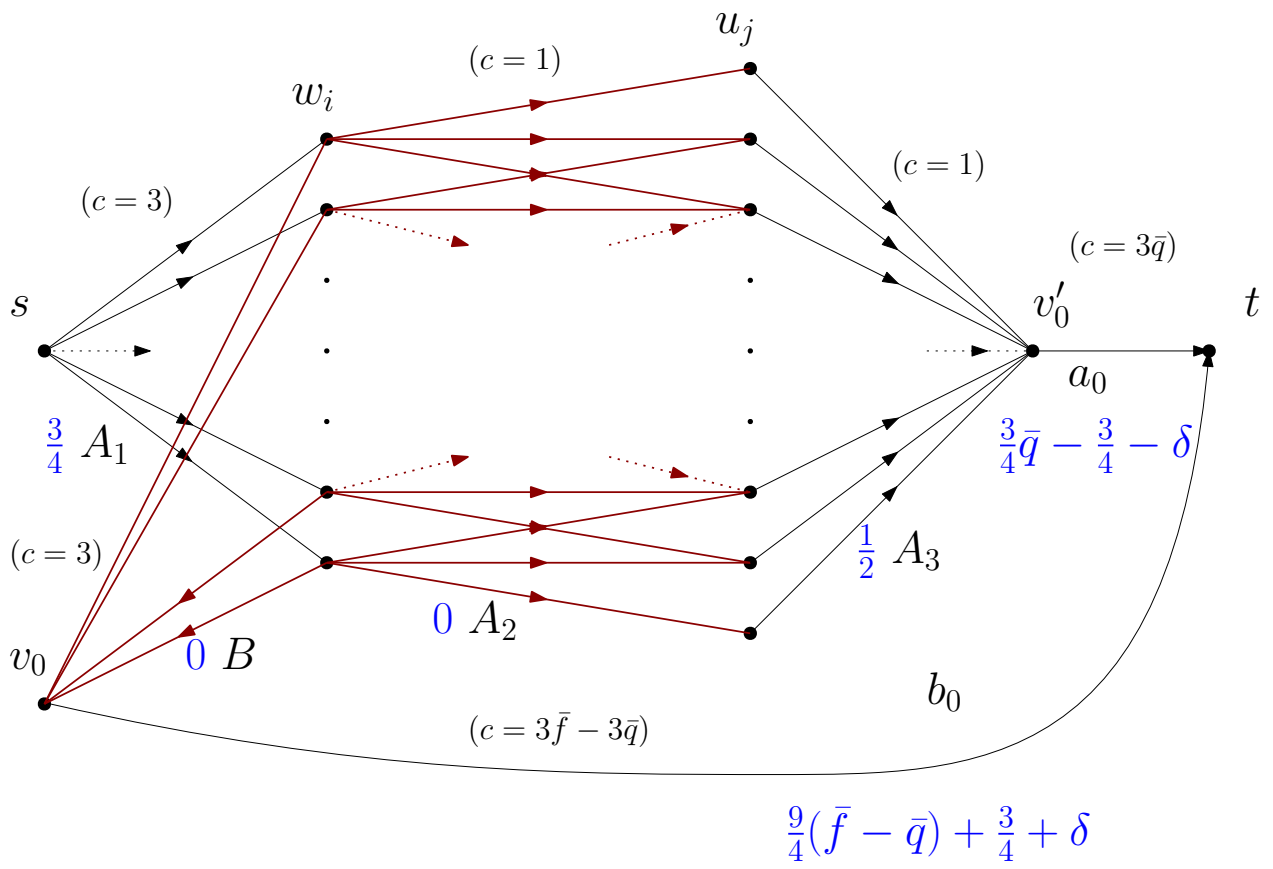
$$x(e) = x(E) - x(E \setminus \{e\}) \leq \nu(E) - \nu(E \setminus \{e\}) = 0.$$

Similarly, $\nu(E \setminus \{e\}) = \nu(E)$ for $e \in B$.

Maschler's Scheme: First Iteration

Lemma (LP_1)

3 $\epsilon_1 = 0, X_1 = C(\Gamma)$ and $\mathcal{J}_1 = \{\{e\} : e \in A_2 \cup B\}$.



Maschler's Scheme: First Iteration

Lemma (LP_1)

(3) $\epsilon_1 = 0, X_1 = C(\Gamma)$ and $\mathcal{J}_1 = \{\{e\} : e \in A_2 \cup B\}$.

Proof: (2) shows $\epsilon_1 = 0, X_1 = C(\Gamma)$. The definition of η_δ and Excess Lemma asserts

$$\forall S \in \mathcal{J} \setminus \{\{e\} : e \in A_2 \cup B\}, \eta_\delta(S) - \nu(S) > 0.$$

Maschler's Scheme: First Iteration

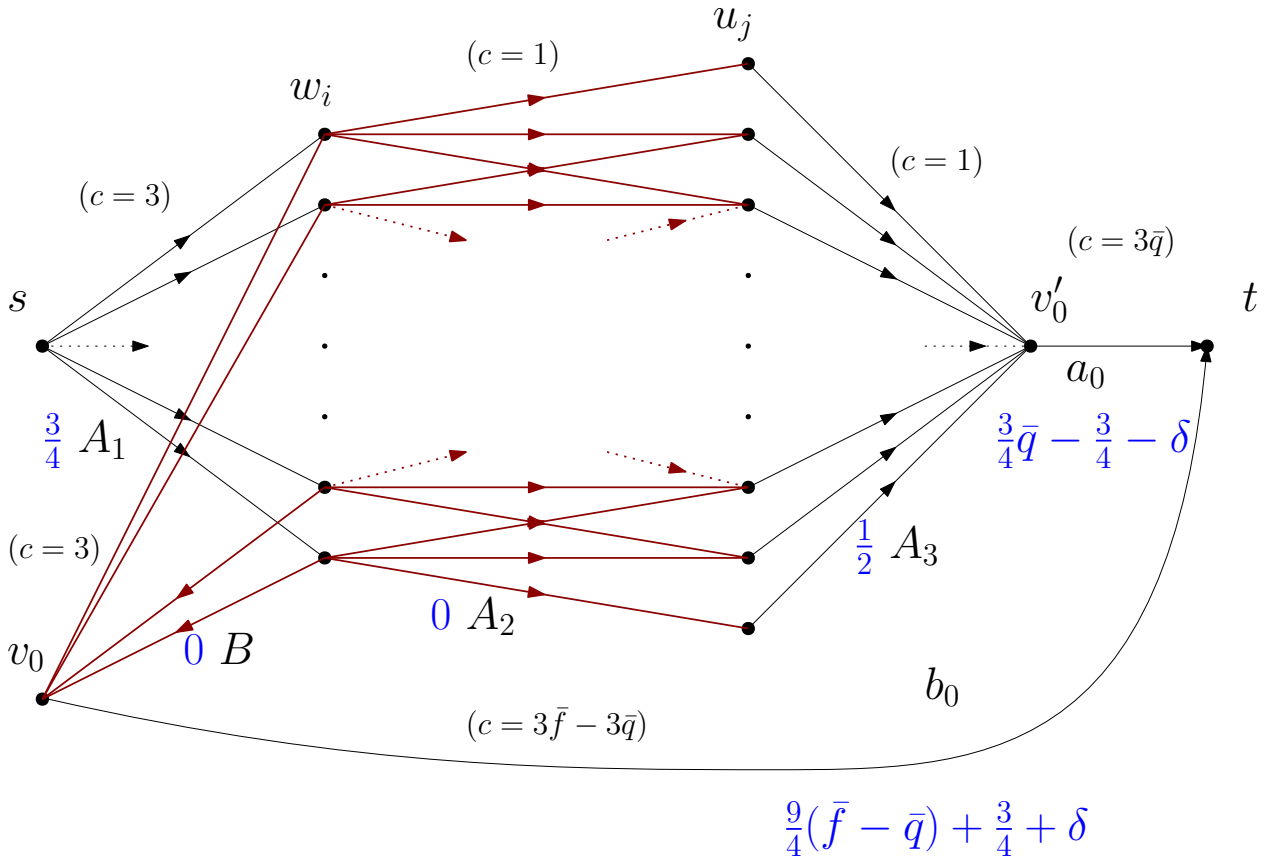
Lemma (LP_1)

1. $\eta_\delta \in C(\Gamma)$.
2. $\forall x \in C(\Gamma), x(e) = 0$ for all $e \in A_2 \cup B$.
3. $\epsilon_1 = 0, X_1 = C(\Gamma)$ and $\mathcal{J}_1 = \{\{e\} : e \in A_2 \cup B\}$.

Maschler's Scheme: Second Iteration

Lemma (LP_2)

1. $\epsilon_2 = \frac{1}{2}$ and η_δ is an optimal solution.



Maschler's Scheme: Second Iteration

Lemma (LP_2)

(1) $\epsilon_2 = \frac{1}{2}$ and η_δ is an optimal solution.

Proof: Let (x, ϵ) be a feasible solution to LP_2. For all $e \in A_3$,

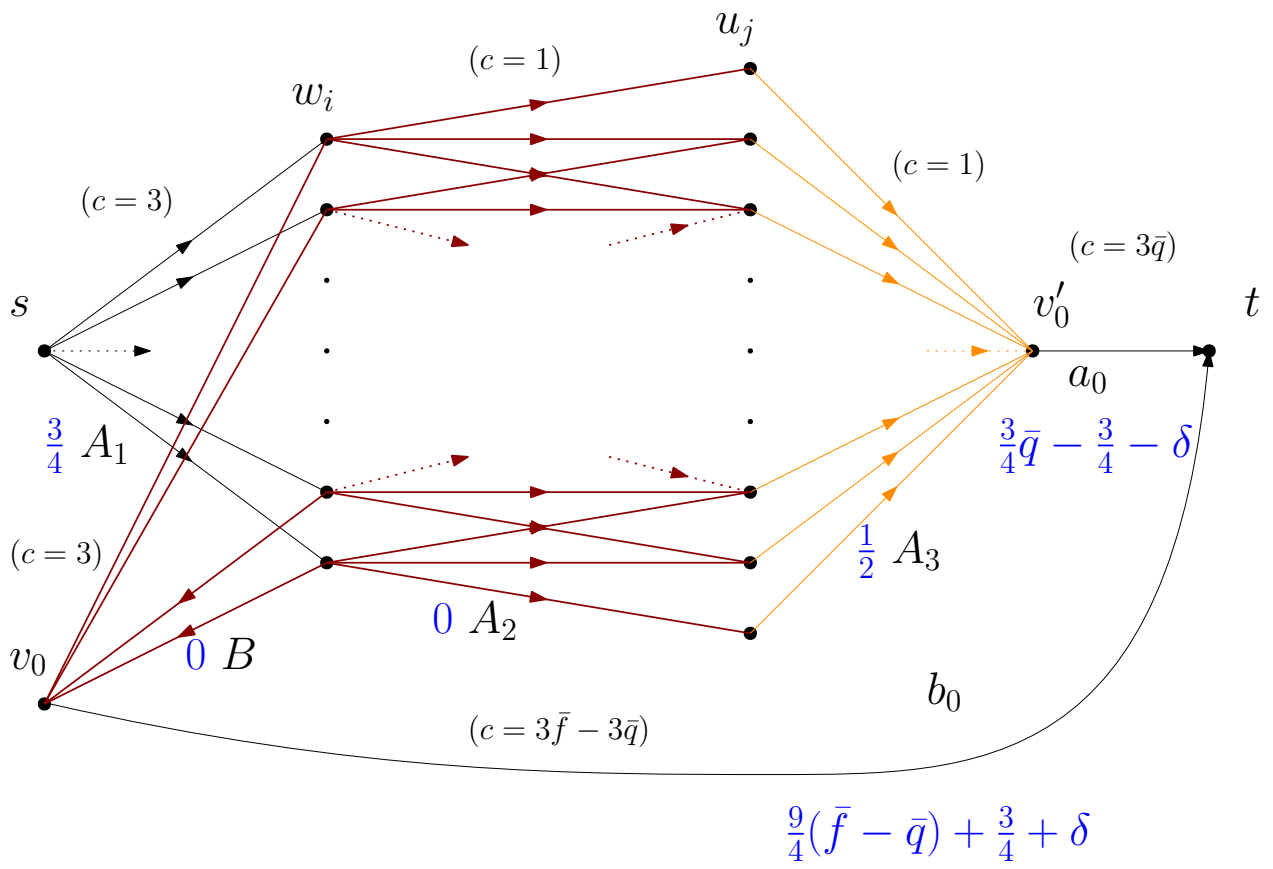
$$x(e) = x(E) - x(E \setminus \{e\}) \leq \nu(E) - (\nu(E) - 1 + \epsilon) = 1 - \epsilon.$$

Thus $\epsilon \leq x(e) \leq 1 - \epsilon$ and $\epsilon \leq \frac{1}{2}$.
By the Excess Lemma, $\eta_\delta(S) - \nu(S) \geq \frac{1}{2}$ for each $S \in \mathcal{J} \setminus \mathcal{J}_1$.
Thus it attains the upper bound.

Maschler's Scheme: Second Iteration

Lemma (LP_2)

❷ $\mathcal{J}_2 = \{\{e\} : e \in A_3\} \cup \{E \setminus \{e\} : e \in A_3\}$.



Maschler's Scheme: Second Iteration

Lemma (LP_2)

2. $\mathcal{J}_2 = \{\{e\} : e \in A_3\} \cup \{E \setminus \{e\} : e \in A_3\}$.

Proof: Let x be optimal in LP_2. Since $x(e) = \frac{1}{2}$ for each $e \in A_3$, for all $S \in \{E \setminus \{e\} : e \in A_3\}$,

$$x(S) - \nu(S) = [x(E) - x(e)] - [\nu(E) - 1] = \frac{1}{2}.$$

Conversely, for $S \notin \{E \setminus \{e\} : e \in A_3\}$ with $\nu(S) > 0$, the Excess Lemma shows

$$\eta_\delta(S) - \nu(S) \geq \frac{3}{4} > \frac{1}{2}.$$

Maschler's Scheme: Second Iteration

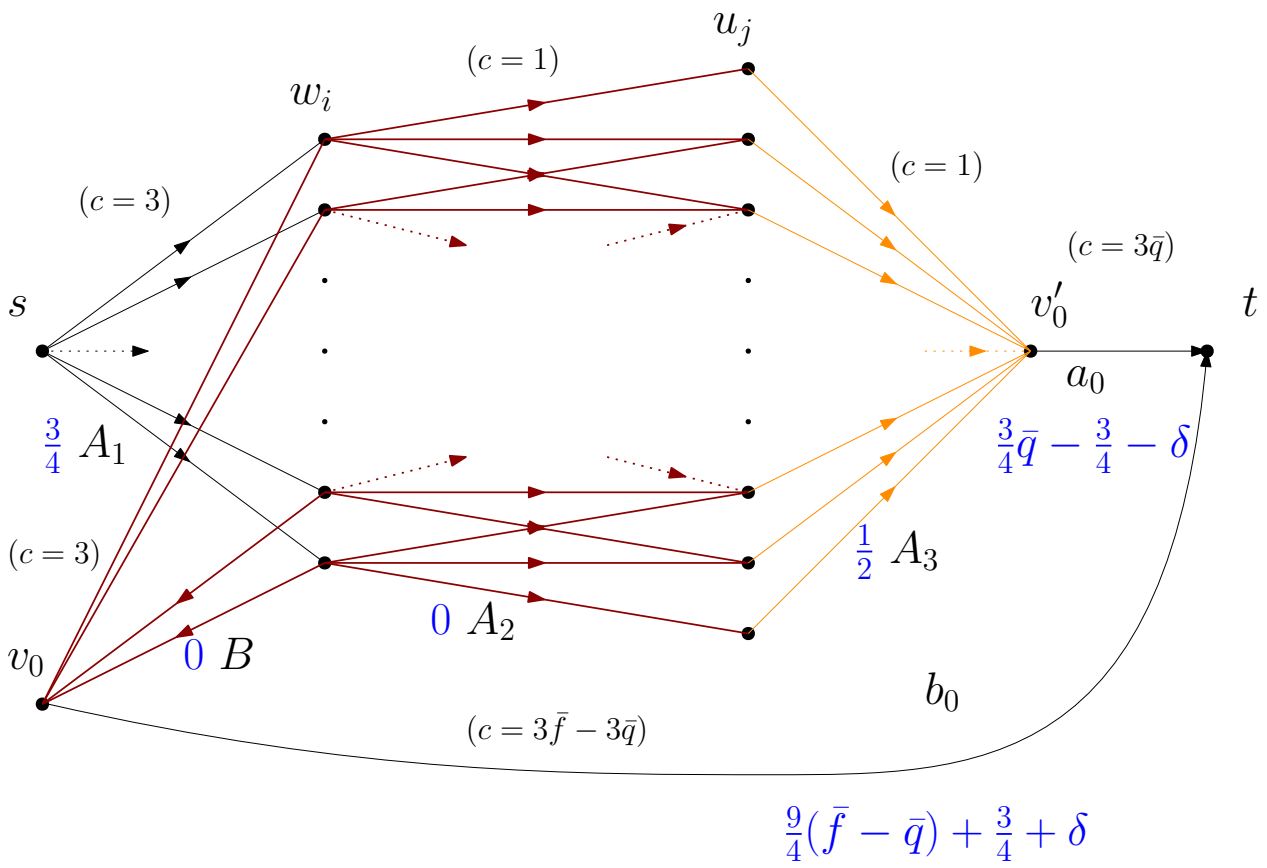
Lemma (LP_2)

1. $\epsilon_2 = \frac{1}{2}$ and η_δ is an optimal solution.
2. $\mathcal{J}_2 = \{\{e\} : e \in A_3\} \cup \{E \setminus \{e\} : e \in A_3\}$.

Maschler's Scheme: Third Iteration

Lemma (LP_3)

1. $\epsilon_3 = \frac{3}{4}$ and η_δ is an optimal solution.



Maschler's Scheme: Third Iteration

Lemma (LP_3)

1. $\epsilon_3 = \frac{3}{4}$ and η_δ is an optimal solution.

Proof: Let (x, ϵ) be a feasible solution to LP_3. Notice it is also optimal in LP_1, LP_2. For all $e \in A_1$,

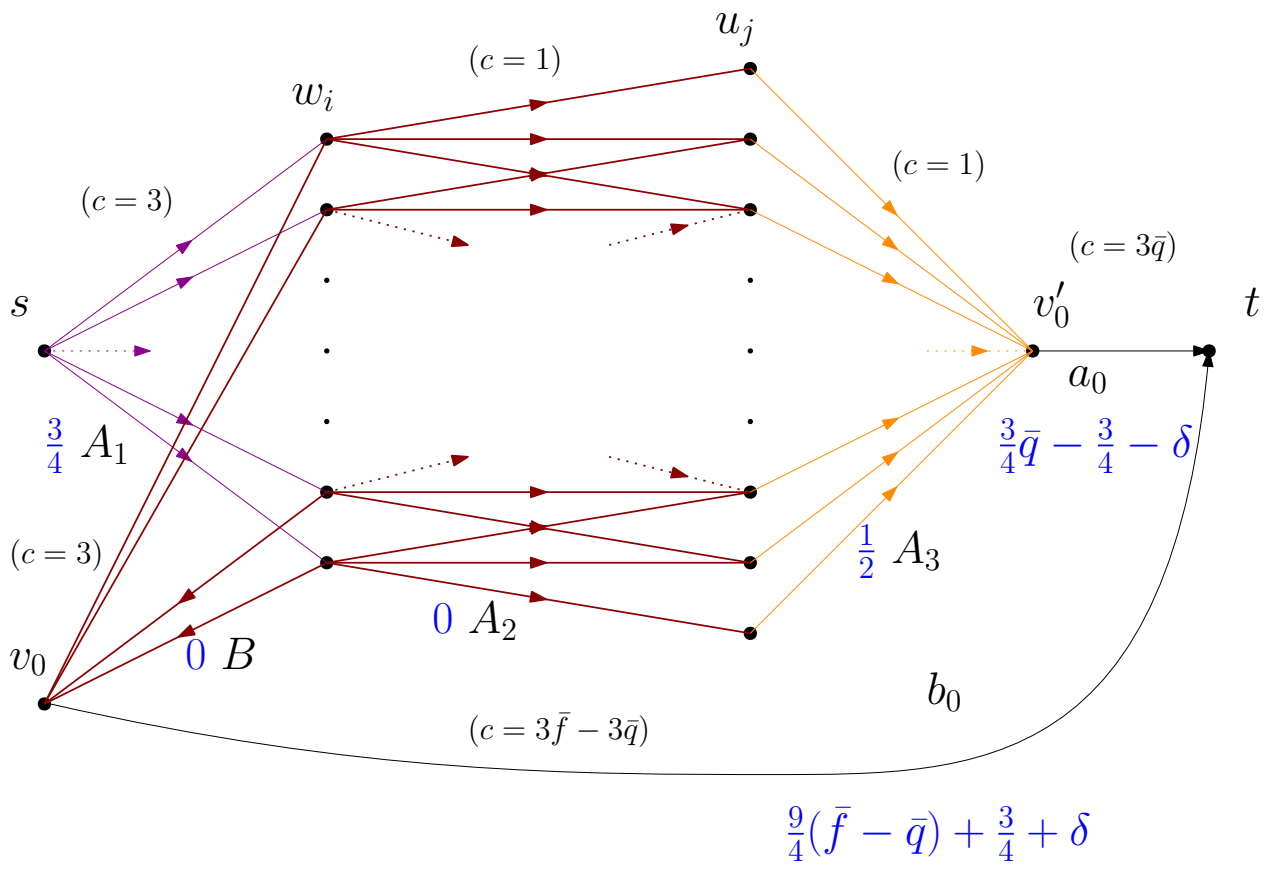
$$x(e) + \frac{3}{2} = x(E) - x(E \setminus T(e)) \leq \nu(E) - [\nu(E) - 3 + \epsilon] = 3 - \epsilon.$$

Thus $\epsilon \leq x(e) \leq \frac{3}{2} - \epsilon$ and $\epsilon \leq \frac{3}{4}$.
By the Excess Lemma, $\eta_\delta(S) - \nu(S) \geq \frac{3}{4}$ for each $S \in \mathcal{J} \setminus (\mathcal{J}_1 \cup \mathcal{J}_2)$. Thus it attains the upper bound.

Maschler's Scheme: Third Iteration

Lemma (LP_3)

❷ $\mathcal{J}_3 \supseteq \{\{e\} : e \in A_1\} \cup \{E \setminus T(e) : e \in A_1\}$.



Maschler's Scheme: Third Iteration

Lemma (LP_3)

❷ $\mathcal{J}_3 \supseteq \{\{e\} : e \in A_1\} \cup \{E \setminus T(e) : e \in A_1\}$.

Proof: Let x be optimal in LP_3. Since $x(e) = \frac{3}{4}$ for each $e \in A_1$, $x(e) = \frac{1}{2}$ for each $e \in A_3$, for all $S \in \{E \setminus T(e) : e \in A_1\}$,

$$x(S) - \nu(S) = [x(E) - x(T(e))] - [\nu(E) - 3] = \frac{3}{4}.$$

Maschler's Scheme: Third Iteration

Lemma (LP_3)

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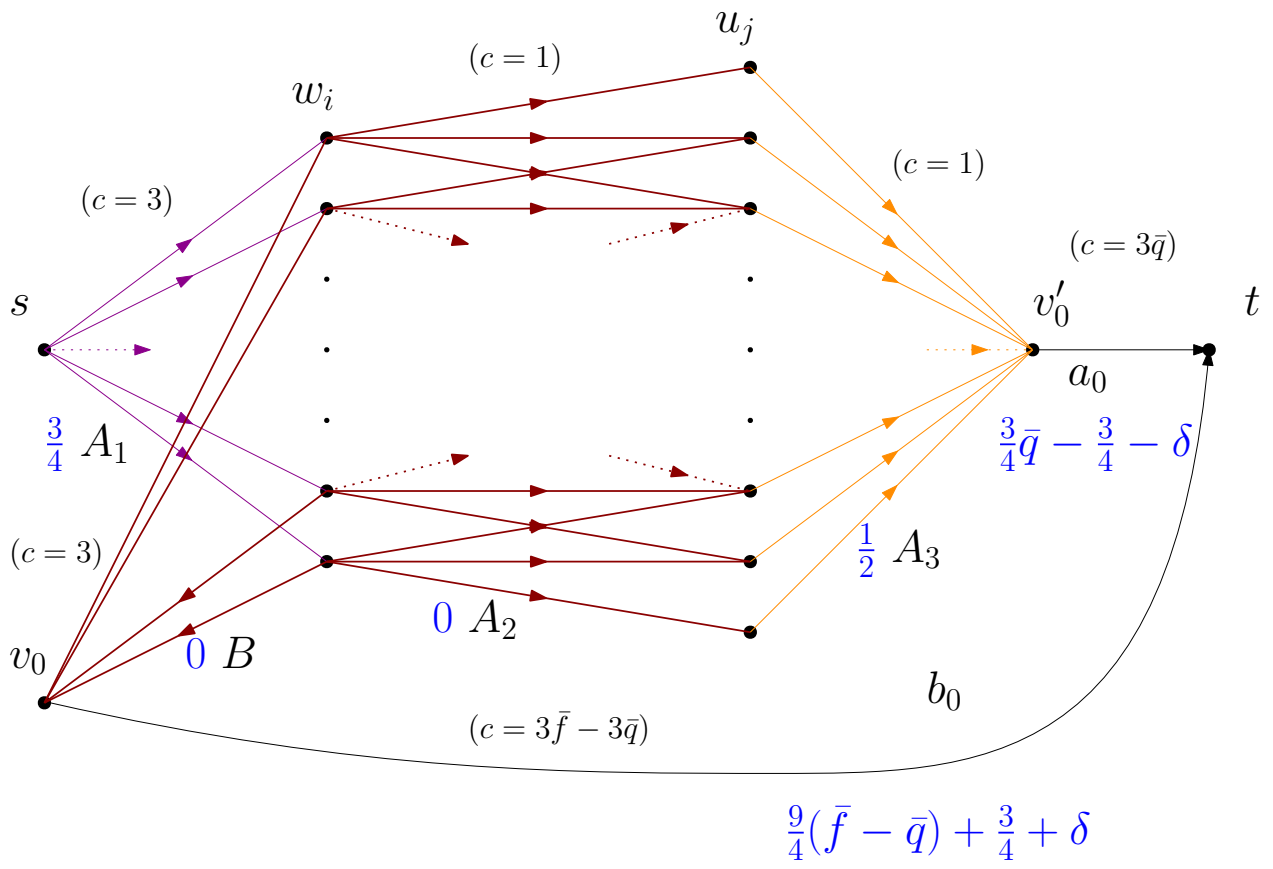
Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If the minimum cover of $\bar{U}$ has size $\bar{q} + 3$, then η_0 is the nucleolus of the flow game Γ.



Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If the minimum cover of $\bar{U}$ has size $\bar{q} + 3$, then η_0 is the nucleolus of the flow game Γ.

Proof (Sketch):

1) Suppose an exact cover of $\bar{U}$ exists and let y be the nucleolus.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If the minimum cover of $\bar{U}$ has size $\bar{q} + 3$, then η_0 is the nucleolus of the flow game Γ.

Proof (Sketch):

1) Suppose an exact cover of $\bar{U}$ exists and let y be the nucleolus.
2) Use the structure of the extra modules and exact cover to construct two disjoint edge coalitions F_1, F_2.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If the minimum cover of $\bar{U}$ has size $\bar{q} + 3$, then η_0 is the nucleolus of the flow game Γ.

Proof (Sketch):

1) Suppose an exact cover of $\bar{U}$ exists and let y be the nucleolus.
2) Use the structure of the extra modules and exact cover to construct two disjoint edge coalitions F_1, F_2.
3) Leverage F_1, F_2 and the optimality of η_0 for LP_3 to determine $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If the minimum cover of $\bar{U}$ has size $\bar{q} + 3$, then η_0 is the nucleolus of the flow game Γ.

Proof (Sketch):

1) Suppose an exact cover of $\bar{U}$ exists and let y be the nucleolus.
2) Use the structure of the extra modules and exact cover to construct two disjoint edge coalitions F_1, F_2.
3) Leverage F_1, F_2 and the optimality of η_0 for LP_3 to determine $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.
4) The lower bounds imply that $3\bar{f} = y(E) = y(a_0) + y(b_0) + \sum_{e \in A_1} y(e) + \sum_{e \in A_3} y(e) \geq 3\bar{f}$.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If an exact cover of $\bar{U}$ exists, $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If an exact cover of $\bar{U}$ exists, $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

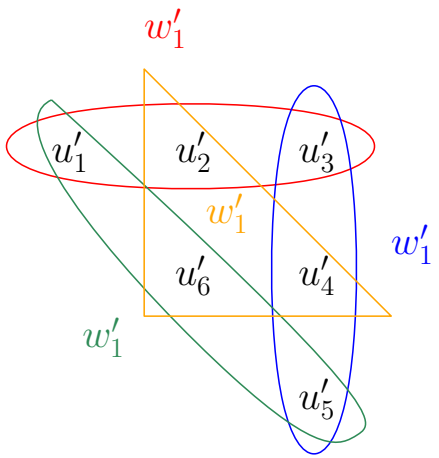
1) Let K be a an exact cover of $\bar{U}$.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If an exact cover of $\bar{U}$ exists, $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

1) Let K be a an exact cover of $\bar{U}$.
2) Choose $u'_i \in U_i$ and $w'_i \in W_i \cap K$ for $i = 1, 2, 3.$ such that $K' := K \setminus \{w'_1, w'_2, w'_3\}$ is a minimum cover of $\bar{U}' := \bar{U} \setminus \{u'_1, u'_2, u'_3\}$.

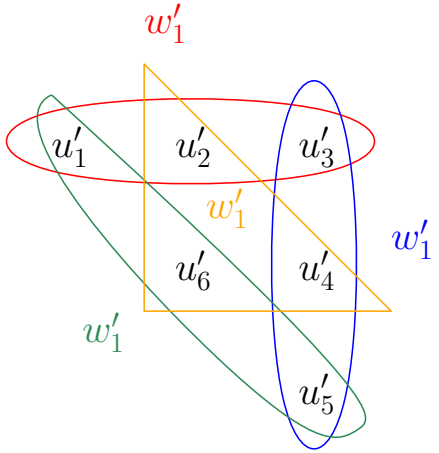


Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If an exact cover of $\bar{U}$ exists, $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

1) Let K be a an exact cover of $\bar{U}$.
2) Choose $u'_i \in U_i$ and $w'_i \in W_i \cap K$ for $i = 1, 2, 3.$ such that $K' := K \setminus \{w'_1, w'_2, w'_3\}$ is a minimum cover of $\bar{U}' := \bar{U} \setminus \{u'_1, u'_2, u'_3\}$.
3) Let $A_1(K) := \{sw : w \in K'\}$ with $|A_1(K)| = \bar{q}$ and $A_3(K) := \{uv'_0 : u \neq u'_i, i = 1, 2, 3\}$ with $|A_3(K)| = 3\bar{q} - 3$.

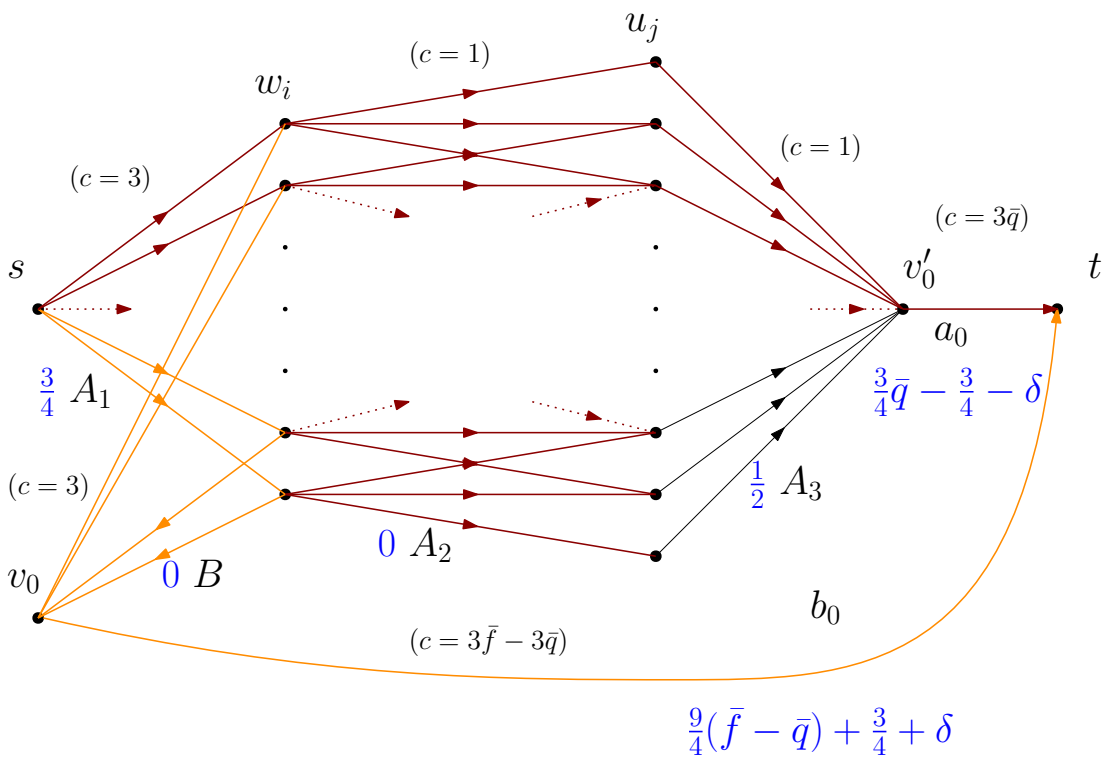


Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If an exact cover of $\bar{U}$ exists, $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

4) Define the disjoint arc subsets
$F_1 = \{a_0\} \cup A_1(K) \cup A_2 \cup A_3(K)$ and
$F_2 = \{b_0\} \cup B \cup (A_1 \setminus A_1(K))$.



Scenario I: Exact Cover Exists

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$F_2 = \{b_0\} \cup B \cup (A_1 \setminus A_1(K))$.

5) $\nu(F_1) = 3\bar{q} - 3$ and $\nu(F_2) = 3(\bar{f} - \bar{q})$.

Scenario I: Exact Cover Exists

Lemma (Nucleolus Characterization I)

If an exact cover of $\bar{U}$ exists, $y(a_0) \geq \eta_0(a_0), y(b_0) \geq \eta_0(b_0)$.

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$F_1 = \{a_0\} \cup A_1(K) \cup A_2 \cup A_3(K)$ and
$F_2 = \{b_0\} \cup B \cup (A_1 \setminus A_1(K))$.

5) $\nu(F_1) = 3\bar{q} - 3$ and $\nu(F_2) = 3(\bar{f} - \bar{q})$.

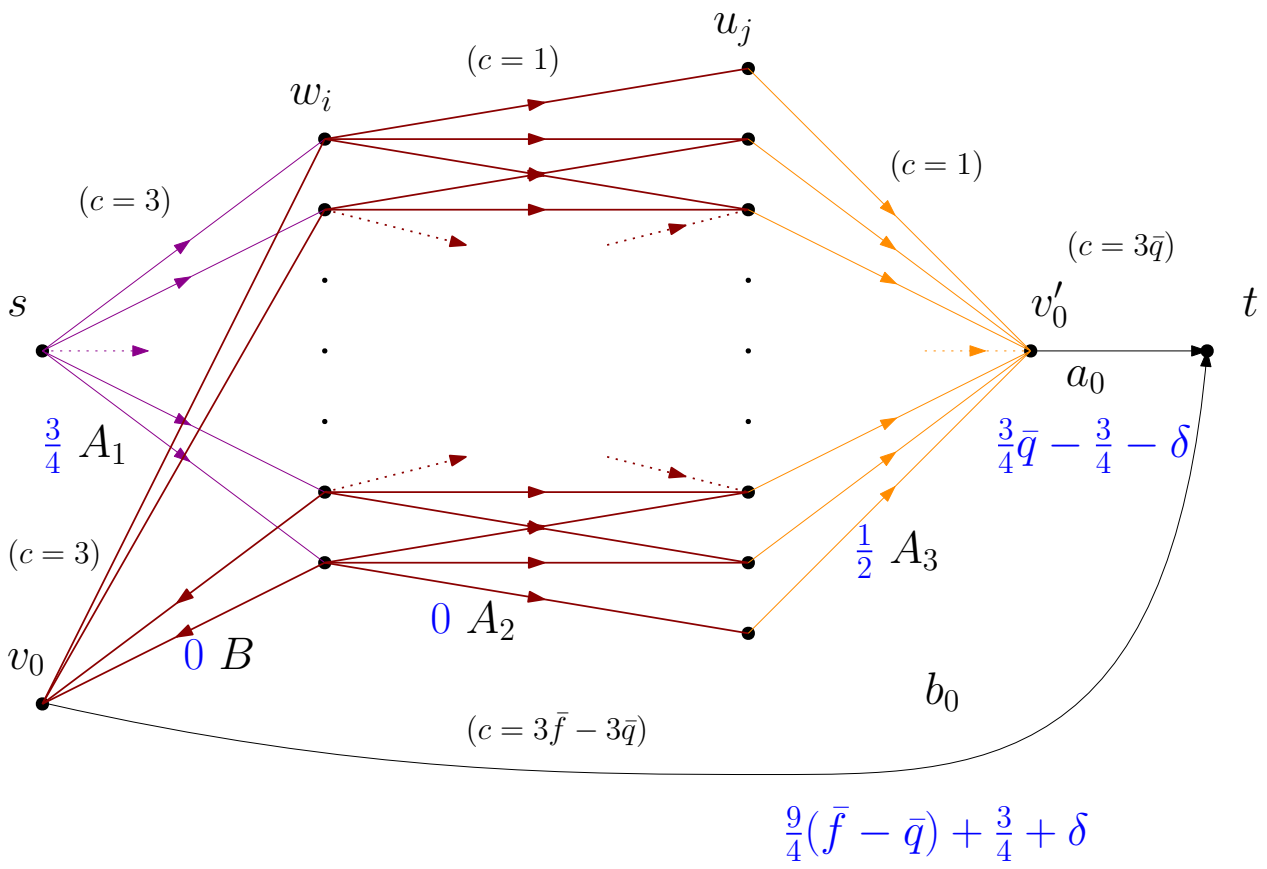
6) Since $y \in \mathrm{OPT}(LP_3)$ and $F_1, F_2 \notin \mathcal{J}_1 \cup \mathcal{J}_2$,

$$\frac{3}{4} \leq y(F_1) - \nu(F_1) = y(a_0) - \frac{3}{4}\bar{q} + \frac{3}{2}$$
$$\frac{3}{4} \leq y(F_2) - \nu(F_2) = y(b_0) - \frac{9}{4}(\bar{f} - \bar{q})$$

Scenario II: No Exact Cover

Lemma (Nucleolus Characterization II)

Pick $0 < \delta < \frac{1}{4}$. If the minimum cover of $\bar{U}$ has size $\bar{q} + 4$, then $\Theta(\eta_\delta)$ is lexicographically greater than $\Theta(\eta_0)$.



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Proof (Sketch):

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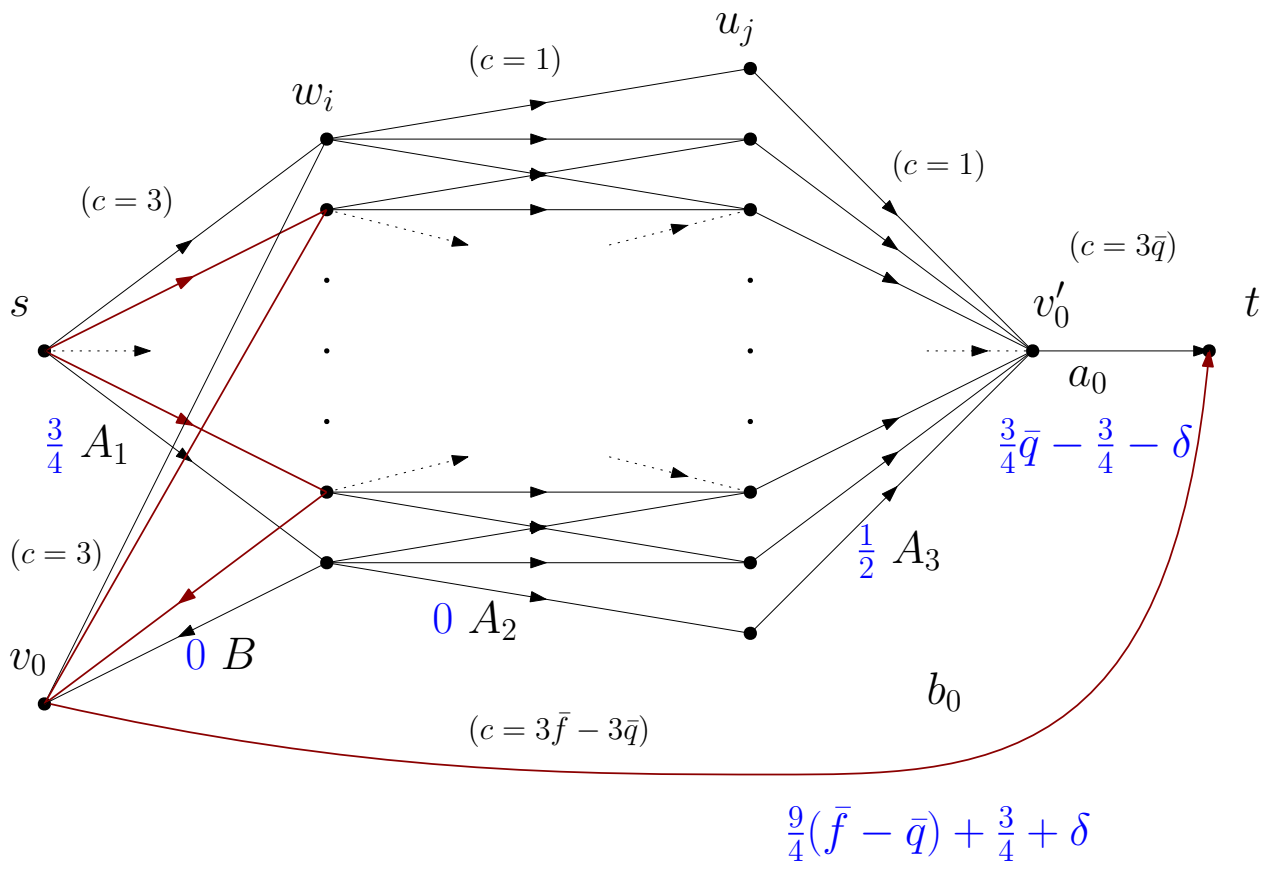

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Conclusion

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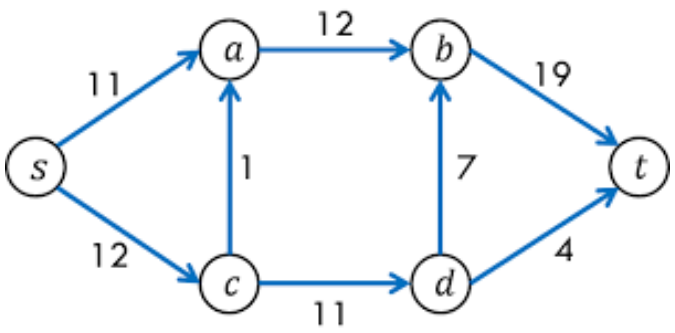

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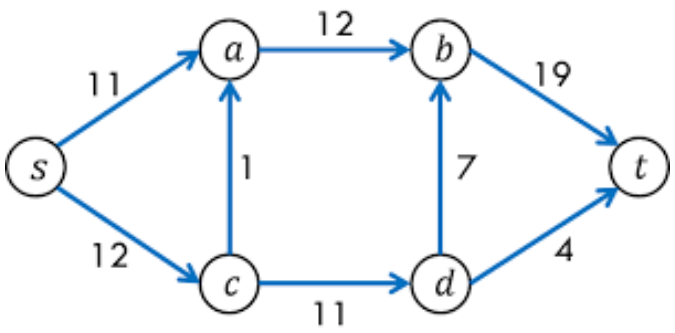

Thanks!